Advanced Differential Equations

Theory of Initial & Boundary Value Problems

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For lots of people

Acknowledgements

I would like to thank everyone.

Preface

I found my first course in differential equations quite boring. Classifying types of differential equations and learning algorithms for generating closed form expressions of their solutions lacked the creativity and problem solving I saw in other areas of math. This book is written to dispell that myth, that an interested undergraduate or beginning graduate student might see that the field of differential equations is interesting, connected broadly to other areas of mathematics, and worth thinking about deeply.

This book is written to be understood by undergraduate students. The only firm prerequisites are multivariable calculus and linear algebra, though experience with proofs, particularly proofs appearing in real analysis, would be helpful. Examples may, from time to time, draw connections to other areas of mathematics which you have not yet studied.

Unlike many differential equations texts written for undergraduates, we will study differential equations as they are studied by mathematicians. While some applications may be discussed, the primary motivation for studying differential equations will be for their own interesting mathematical properties.

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Chapter 1

Initial Value Problems

In this chapter, we will recount some results dealing with initial value problems which you are likely to have encountered in a first course in differential equations. As happens in math, the very powerful theorems we are able to prove about the existence and uniqueness of solutions to initial value problems make them rather dull to study. Nevertheless, we will take advantage of their nice properties later own when studying more difficult types of problems.

1.1 Motivating Examples

In a first course in differential equations, the question of the existence of a solution is often moot, because one can demonstrate the existence simply by writing down the solution.

Example 1.1.1 Show that the differential equation

$$\begin{cases} y'' = y \\ y(0) = 0 \\ y'(0) = 3 \end{cases}$$

has a solution.

Solution. Since the differential equation $y'' = y$ is linear with constant coefficients, we could consider its characteristic equation, find the roots, and use those to write down a solution. The initial conditions would then be used to find unknown constants.

For this problem, however, such a method is likely unnecessary. We may observe that we know two linearly independent functions, namely $y_1(x) = \sin(x)$ and $y_2(x) = \cos(x)$, which satisfy $y'' = y$. Hence, any solution must be of the form $y(t) = c_1 y_1(t) + c_2 y_2(t)$, and the initial conditions are again used to find the unknown constants c_1 and c_2.

In this case, the reader should find that $y(t) = 3\sin(t)$ is a solution. What is harder to show is that this is the *only* solution. □

While the existence of a solution can be demonstrated by simply writing down a candidate and showing that it satisfies the initial value problem, it is harder to know whether the solution you have found is the *only* solution. In a first course in differential equations, you may have enountered such problems.

Example 1.1.2 Show that the differential equation

$$\begin{cases} y' = y^{\frac{2}{3}} \\ y(0) = 0 \end{cases}$$

has two solutions.

Solution. This is a classic example of a separable differential equation. To find the solution, rewrite the differential equation as

$$\frac{y'}{y^{\frac{2}{3}}} = 1, \tag{1.1.1}$$

which is only valid if $y \neq 0$. Note that $y_1(x) = 0$ is indeed a solution to the differential equation.

If $y \neq 0$, then we may integrate both sides of (1.1.1) with respect to x from 0 to t. Then on the left hand side we have

$$\int_0^t \frac{y'}{y^{\frac{2}{3}}}\, dx = \int_{y_2(0)}^{y_2(t)} \frac{1}{y^{\frac{2}{3}}}\, dy = 3(y(t))^{\frac{1}{3}}\Big|_{y_2(0)}^{y_2(t)} = 3(y_2(t))^{\frac{1}{3}},$$

while on the right hand side, we have

$$\int_0^t 1\, dx = t.$$

Hence, our solution is $3(y_2(t))^{\frac{1}{3}} = t$, or equivalently $y_2(t) = \frac{1}{27}t^3$. □

From these two examples, we see that there are three important questions you might ask about an initial value problem:

1. Does the IVP have a solution? This is the question of ***existence***.
2. If the IVP has a solution, is there only one solution? This is the question of ***uniqueness***.
3. If the IVP has a solution which is not unique, how many solutions are there? This is the question of ***multiplicity***.

From our examples thus far, however, we have answered these questions only partially. In Example 1.1.1, we proved existence of a solution, but did not establish uniqueness. In Example 1.1.2, we proved nonuniqueness, but did not definitely answer the question of multiplicity (though we did provide a lower bound on the number of solutions).

In any event, we are in a considerable amount of trouble if our only methods for dealing with these questions rely on writing down explicit solutions. In fact, even simple differential equations of the form

$$y'' + a(t)y' + b(t)y = 0$$

may not have solutions expressible in terms of elementary (trigonometric, exponential, logarithmic, polynomial, etc.) functions. The interested reader is directed to [1] for an in-depth discussion of when such equations do and do not have elementary solutions.

Moving forward, we wish to address the questions of existence, uniqueness, and multiplicity for initial value problems first, and then subsequently for a new type of problem, called a boundary value problem.

1.2 Existence Theorems

Definition 1.2.1 A function $\vec{f} : D \to \mathbb{R}^n$ is said to satisfy a *Lipschitz condition with respect to* x on the open set $D \subset \mathbb{R} \times \mathbb{R}^n$ if, for any rectangle

$$\Omega := \{(t, \vec{x}) \mid |t - t_0| \leq a,\ \|\vec{x} - \vec{x}_0\| \leq b\} \subset D,$$

there exists a constant K (depending on Ω) such that

$$\|\vec{f}(t, \vec{x}) - \vec{f}(t, \vec{y})\| \leq K\|\vec{x} - \vec{y}\|$$

for all $(t, \vec{x}), (t, \vec{y}) \in \Omega$. $\Diamond$

Theorem 1.2.2 *Consider the differential equation*

$$\begin{cases} \vec{x}' = \vec{f}(t, \vec{x}) \\ \vec{x}(\alpha) = \vec{\beta} \end{cases} \tag{1.2.1}$$

where $\alpha \in \mathbb{R}$, $\vec{\beta} \in \mathbb{R}^n$, *and* $\vec{f} : D \to \mathbb{R}^n$ *is a continuous function defined on an open set* $D \subset \mathbb{R}^{n+1}$ *containing* $(\alpha, \vec{\beta})$.

If $\vec{f}$ *satisfies a Lipschitz condition with respect to* x, *then (1.2.1) has a unique solution in a neighborhood of* $t = \alpha$.

1.3 My Exercises

1. Is this an exercise?

Answer. Yes.

Solution. It is, but not a very good one.

2. Another, with a hint.

Hint. The hint.

Chapter 2

(No title)

Try adding your own content here!

Chapter 3

Examples of PreTeXt features

Below are examples of a lot of PreTeXt elements.

3.1 Environments and Blocks

Some text

Theorem 3.1.1 My Theorem. *Theorem statement.*

Proof. Proof of theorem. ■

Example 3.1.2 Statement of example

Solution. The solution. □

Now a figure. (Uncomment the source to include Tikz, SagePlot, and Asymptote examples built with `pretext build -d`.)

Figure 3.1.3 A frog

3.2 Another section

This will have more stuff

Appendix A

Selected Hints

1 · Initial Value Problems
1.3 · My Exercises

1.3.2.

The hint.

Appendix B

Selected Solutions

1 · Initial Value Problems
1.3 · My Exercises

1.3.1.

It is, but not a very good one.

Appendix C

List of Symbols

Symbol	Description	Page

References

[1] Kovacic, J.J., *An algorithm for solving second order linear homogeneous differential equations.*, Journal of Symbolic Computation; (1986) **2** no. 1 3-43.

Colophon

This book was authored in PreTeXt.